1. Create `main.c` that accepts two integers and prints addition. Write a Makefile with targets:

- `all` (default) → builds `app`
- `run` → runs `./app`
- `clean` → removes binary

Deliverable: `app`, `run`, `clean`

2. Rewrite Q1 Makefile using:

- `CC`, `CFLAGS`, `TARGET`, `SRC`

Use `CFLAGS = -Wall -Wextra -O2`

Deliverable: variables + working build

3. Add a `help` target that prints usage:

- how to build
- how to run
- how to clean

Use `@echo` lines.

Deliverable: `make help` shows instructions.

4. Two Files Addition

Create:

- `main.c` (takes input, calls function)
- `helper.c` (implements `Add`)
- `helper.h`

Makefile should generate:

- `main.o`, `helper.o`
- link into `app`

Deliverable: correct dependency on `helper.h`

5. Static Library Build (`.a`)

Create:

- `libmath.a` containing `Add`, `Sub`, `Mul`, `Div`
  Make targets:
- `lib` → builds library
- `app` → links app using static library
- `clean`

Deliverable: `ar rcs` usage inside Makefile

6. Shared Library Build (`.so`)

Create:

- `libmath.so` (PIC code)
  Use flags:
- `-fPIC`
- `-shared`

Make targets:

- `so`
- `app`
- `run` (must set `LD_LIBRARY_PATH=.`)

Deliverable: working shared library app